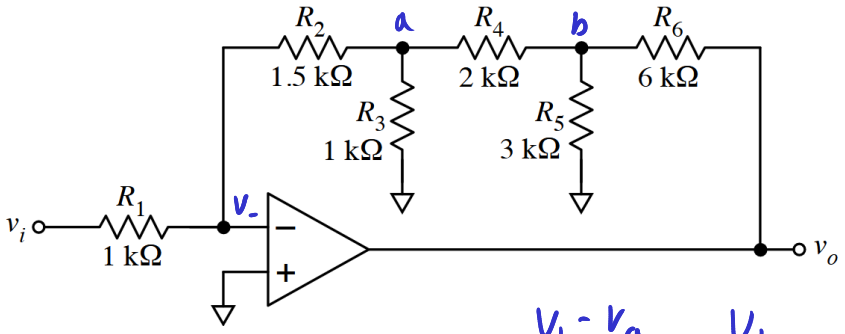
EE 2010
Fall 2026

Name _____

Game 37
10/18/2026

Find the gain of the op-amp circuit below



$V_- = V_+ = 0V$

$\frac{V_a}{1.5} + \frac{V_a}{1} + \frac{V_a - V_b}{2} = 0$

$4V_a + 6V_a + 3V_a - 3V_b = 0$

$13V_a = 3V_b$

$V_b = \frac{13}{3} V_a$

$\frac{V_b - V_a}{2} + \frac{V_b}{3} + \frac{V_b - V_o}{6} = 0$

$3V_b - 3V_a + 2V_b + V_b - V_o = 0$

$6V_b - 3V_a = V_o$

$26V_a - 3V_a = V_o$

$23V_a = V_o$

$V_a = \frac{1}{23} V_o$

$\frac{-V_i}{1} - \frac{V_a}{1.5} = 0$

$-3V_i - 2V_a = 0$

$V_a = -\frac{3}{2} V_i$

$\frac{1}{23} V_o = -\frac{3}{2} V_i$

$V_o = -\frac{69}{2} V_i$

$\text{gain} = \frac{V_o}{V_i} = -\frac{69}{2} = -34.5V$